AN ASYMMETRIC DYNASTI ANALYSIS

Asymmetric Inertia in Momentary Affect and Its Links to Sleep Quality

Ningzhe Zhu[1*] · Ramona Schoedel[1,2] · Larissa Sust[1] · Markus Bühner[1,3] · Ai Ye[1,3]

[1] Department of Psychology, LMU Munich · [2] Charlotte Fresenius Hochschule · [3] Max Planck Institute for Human Development

*Correspondence Ningzhe Zhu: ningzhe.zhu@lmu.de

1 INTRODUCTION

- Conventional autoregressive and cross-lagged models impose one temporal coefficient across the full predictor range.
- This assumes above-usual and below-usual states carry over with equal strength.
- DYNamics of ASymmetric TIme series (DYNASTI) estimates regime-specific paths around each person's own mean.

2 METHODS

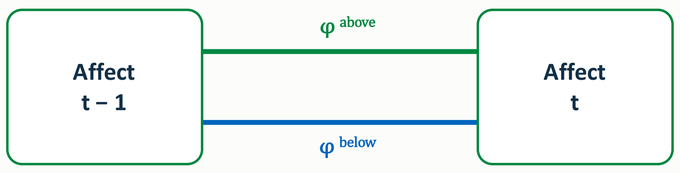
STUDY 1 | MOMENTARY AFFECT

N = 434 • 9,424 EMA observations • up to 4 prompts/day × 12 days

Bayesian DSEM | asymmetric AR(1)

$$x_{it} = \varphi_i^{above}\, x_{i,t-1} + \varepsilon_{it} \quad \text{if } x_{i,t-1} > 0$$
$$x_{it} = \varphi_i^{below}\, x_{i,t-1} + \varepsilon_{it} \quad \text{if } x_{i,t-1} \le 0$$

x_{it} = affect deviation from participant i's mean. Missing EMA values were jointly estimated.



The same transition is estimated separately by threshold regime.

STUDY 2 | DAILY AFFECT ↔ SLEEP

N = 425 • up to 12 daily observations • separate model for each affect

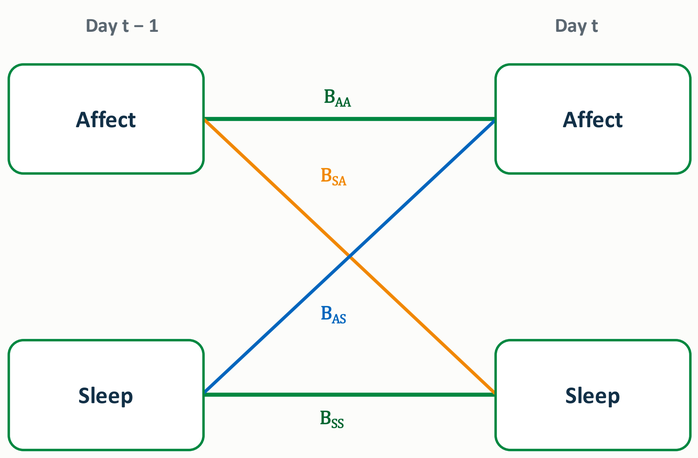
Bayesian RI-CLPM | four regime-specific lagged paths

$$A_{it} = \mu A + \eta_i A + B_{AA}(dA_{i,t-1}) + B_{AS}(dS_{i,t-1}) + \varepsilon A_{it}$$
$$S_{it} = \mu S + \eta_i S + B_{SS}(dS_{i,t-1}) + B_{SA}(dA_{i,t-1}) + \varepsilon S_{it}$$

$$B(d) = \beta^{above}\, d \cdot I(d > 0) + \beta^{below}\, d \cdot I(d \le 0)$$

$$I(\text{condition}) = \begin{cases} 1, & \text{if the condition is true,} \\ 0, & \text{if the condition is false.} \end{cases}$$

A = daily affect, S = prior-night sleep quality, η = person random intercept. Each path is selected by its own lagged predictor.



Within-person autoregressive and reciprocal cross-lagged paths

3 RESULTS

STUDY 1 | THRESHOLD DYNAMICS

Slopes seem different between cases when the deviations at t-1 is larger than 0 and other cases when the deviation at t-1 is smaller than 0.



Values are person–centered. Points are equal–frequency bins within each threshold regime; point size represents the number of transitions.



* Credible above–below difference. After adjustment, arousal remained credibly asymmetric.

STUDY 2 | RI-CLPM PATH RESULTS

Four within-person paths were estimated separately by the regime of each lagged predictor.

Within-person path	Valence ↔ Sleep	Stress ↔ Sleep	Arousal ↔ Sleep
$Affect_{t-1} \rightarrow Affect_t$	Above .200 | Below .158	Above .187 | Below .209	Above .087 | Below .204
$Sleep_{t-1} \rightarrow Sleep_t$	Above .256 | Below −.016 Δ +.270*	Above .263 | Below −.020 Δ +.282*	Above .296 | Below −.003 Δ +.330*
$Sleep_{t-1} \rightarrow Affect_t$	Above .019 | Below −.005	Above −.073 | Below +.060 Δ −.133*	Above .037 | Below .039
$Affect_{t-1} \rightarrow Sleep_t$	Above −.002 | Below .066	Above −.062 | Below −.011	Above −.019 | Below .072

KEY PATTERN Better-than-usual sleep persisted across all models; below-usual sleep did not. Only Sleep → Stress showed credible cross-lagged asymmetry.

4 DISCUSSION

MAIN FINDINGS

- Valence and arousal showed stronger momentary persistence above baseline.
- Better-than-usual sleep showed stronger next-day carryover.
- Among well-slept days, better sleep predicts lower stress, whereas on other days, better sleep predicts higher stress.

INTERPRETATION

- Inertia is state-dependent—not a uniform index of emotional rigidity.
- Pleasant-state maintenance and recovery from unfavorable states may reflect different regulatory processes.
- The prediction of sleep on next-day stress is context-dependent.

Values are posterior means; intervals are 95% HDI / credible intervals. * interval excludes zero.

Schaaf, J. V., Sørensen, Ø., McCormick, E. M., Aristodemou, M., & Kievit, R. A. (2025). Uncovering Asymmetric Temporal Dynamics Using Threshold Dynamics Parameters. *Structural Equation Modeling: A Multidisciplinary Journal*, *32*(6), 949–965. https://doi.org/10.1080/10705511.2025.2519208